10.3 The Lorentz Group: Components, Algebra and Unitarity

Conventions. As in Srednicki: metric $g = \mathrm{diag}(-1, +1, +1, +1)$, and a Lorentz transformation near the identity is represented by $U(1 + \delta\omega) = I + \frac{i}{2}\delta\omega_{\mu\nu}M^{\mu\nu}$. Rotation and boost generators are

$$J_i = \tfrac{1}{2}\epsilon_{ijk}M^{jk}, \qquad K_i = M^{0i}.$$

Srednicki defines $K_i = M^{i0}$, the negative of ours; none of the commutators below depends on that sign, and our N_i, $\tilde{N}_i$ (§10.20) coincide with his N_i, $N_i^\dagger$.

Connected Components

The Lorentz group $O(1,3)$, which preserves the spacetime interval $ds^2 = -dt^2 + dx^2 + dy^2 + dz^2$, is a **disconnected** Lie group. It splits into four disjoint subsets called **connected components**, which are distinguished by the discrete symmetries of parity (P) and time reversal (T).
A **connected component** of a topological space is a maximal connected subset:

- **Connected**: no separation into two disjoint non-empty open sets.
- **Maximal**: not contained in any larger connected subset.

Even if the entire space is disconnected, its connected components are the "largest possible" internally connected pieces. The Lorentz group decomposes into:

- $\mathbf{SO^+(1,3)}$: proper orthochronous transformations ($\det = +1$, preserves time orientation). This is the identity-connected subgroup.
- $\mathbf{T \cdot SO^+(1,3)}$: proper non-orthochronous ($\det = +1$, reverses time).
- $\mathbf{P \cdot SO^+(1,3)}$: improper orthochronous ($\det = -1$, preserves time).
- $\mathbf{PT \cdot SO^+(1,3)}$: improper non-orthochronous ($\det = -1$, reverses time).

Key properties:

- **Disjoint & disconnected**: no continuous path exists between distinct components.
- **Internally connected**: within each component, any two transformations can be linked by a continuous path (e.g., via boosts or rotations).

- **Discrete symmetries**: P and T map between components. For example, $P \cdot SO^+(1,3)$ is the parity-transformed copy of $SO^+(1,3)$.
- **Standard terminology**: the four pieces partition $O(1,3)$, just as $O(n)$ has two components, $SO(n)$ and $O(n) \setminus SO(n)$.

Physical significance. Only $SO^+(1,3)$ is an exact symmetry of nature: the weak interaction violates P, and T (equivalently CP), while the combination CPT is conserved. The other three components are therefore not symmetries by themselves, which is why the discrete transformations are treated separately from continuous Lorentz invariance.

The Lorentz Algebra

The generators $M^{\mu\nu} = -M^{\nu\mu}$ satisfy

$$[M^{\mu\nu}, M^{\rho\sigma}] = i\left(g^{\mu\rho}M^{\nu\sigma} - g^{\nu\rho}M^{\mu\sigma} - g^{\mu\sigma}M^{\nu\rho} + g^{\nu\sigma}M^{\mu\rho}\right).$$

In terms of rotations and boosts:

$$[J_i, J_j] = i\epsilon_{ijk}J_k, \qquad [J_i, K_j] = i\epsilon_{ijk}K_k, \qquad [K_i, K_j] = -i\epsilon_{ijk}J_k.$$

The minus sign in the last commutator is the defining feature of the Minkowskian algebra: two successive boosts in different directions are a boost *and* a rotation (the Thomas–Wigner rotation). Its consequences for spinors are worked out in §10.20.

Generators on Fields

Under $U(\Lambda)$, fields transform as $U(\Lambda)^{-1}\Phi_A(x)U(\Lambda) = D_A{}^B(\Lambda)\,\Phi_B(\Lambda^{-1}x)$, with $D(1+\delta\omega) = I + \frac{i}{2}\delta\omega_{\mu\nu}S^{\mu\nu}$. The total generator acting on a field is an orbital part plus a spin part.

- **Scalar fields (spin 0)**: $\phi(x) \to \phi(\Lambda^{-1}x)$, no spin part. In Srednicki's form $[\phi(x), M^{\mu\nu}] = \mathcal{L}^{\mu\nu}\phi(x)$ with the orbital operator

$$\mathcal{L}^{\mu\nu} = \frac{1}{i}\left(x^\mu\partial^\nu - x^\nu\partial^\mu\right).$$

- **Vector fields (spin 1)**: $V^\mu(x) \to \Lambda^\mu{}_\nu V^\nu(\Lambda^{-1}x)$, with the spin part

$$(S_V^{\mu\nu})^\rho{}_\tau = \frac{1}{i}\left(g^{\mu\rho}\delta^\nu{}_\tau - g^{\nu\rho}\delta^\mu{}_\tau\right), \qquad \tfrac{i}{2}\delta\omega_{\mu\nu}(S_V^{\mu\nu})^\rho{}_\tau = \delta\omega^\rho{}_\tau.$$

- **Spinor fields (spin 1/2)**: $\Psi(x) \to D(\Lambda)\Psi(\Lambda^{-1}x)$, with

$$S^{\mu\nu} = \frac{i}{4}[\gamma^\mu, \gamma^\nu], \qquad \{\gamma^\mu, \gamma^\nu\} = -2g^{\mu\nu}.$$

Hermiticity of the Generators in Different Representations

The hermiticity of J_i and K_i depends on the representation, and the pattern explains which representations can be unitary. We also use the complex combinations $N_i = \tfrac{1}{2}(J_i + iK_i)$ and $\tilde{N}_i = \tfrac{1}{2}(J_i - iK_i)$ of §10.20.

The defining (vector) representation. Acting on 4-vectors, the generators are 4×4 matrices. For example, the rotation generator $J_z = S^{12}_V$ and boost generator $K_x = S^{01}_V$ are

$$J_z = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -i & 0 \\ 0 & i & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad K_x = \begin{pmatrix} 0 & i & 0 & 0 \\ i & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Rotation generators are Hermitian, boost generators anti-Hermitian, and N_i, $\tilde{N}_i$ are both Hermitian. This representation describes how 4-vectors (positions, momenta, the electromagnetic potential) transform; it acts on the components of a field, not on the quantum states, so its non-unitarity is harmless.

The spinor representation. From $S^{\mu\nu}$ above,

$$J_i = \frac{i}{4}\epsilon_{ijk}\gamma^j\gamma^k, \qquad K_i = \frac{i}{2}\gamma^0\gamma^i.$$

In the Weyl basis these are $J_i = \tfrac{1}{2}\,\mathrm{diag}(\sigma_i, \sigma_i)$ and $K_i = \tfrac{i}{2}\,\mathrm{diag}(-\sigma_i, \sigma_i)$: J_i Hermitian, K_i anti-Hermitian, and N_i, $\tilde{N}_i$ Hermitian. The upper and lower blocks are the left-handed $(2,1)$ and right-handed $(1,2)$ Weyl spinors of §10.20. The tilde in $\tilde{N}_i$ is a label for the second $\mathfrak{su}(2)$, not a hermitian conjugate.

Unitary representations on states. On the Hilbert space of physical states the representation must be unitary, so all six generators are Hermitian, $J_i^\dagger = J_i$, $K_i^\dagger = K_i$, and then $N_i^\dagger = \tilde{N}_i$. These representations are necessarily infinite-dimensional.

Property	**Vector**	**Dirac spinor**	**States**
Space	$\mathbb{C}^4$ (components)	$\mathbb{C}^4$ (components)	one-particle Hilbert space
Dimension	4	4	infinite
J_i	Hermitian	Hermitian	Hermitian
K_i	anti-Hermitian	anti-Hermitian	Hermitian
N_i, $\tilde{N}_i$	both Hermitian	both Hermitian	$N_i^\dagger = \tilde{N}_i$
Unitary?	No	No	Yes

Lorentz generators in three representations

Finite-Dimensional Representations Are Not Unitary

The Lorentz group is **non-compact**: boosts generate transformations with unbounded rapidity. A non-compact simple Lie group has **no non-trivial finite-dimensional unitary representations**. That is why the field components above (vectors, spinors) transform non-unitarily, while the states must and do transform under **infinite-dimensional unitary** representations. Quantum field theory needs both: fields carry finite-dimensional non-unitary representations, which make covariance manifest, and states carry infinite-dimensional unitary ones, which conserve probability.

Unitary Representations of the Poincaré Group

Unitary representations are built on the states, following Wigner's classification of the irreducible unitary representations of the Poincaré group.

Massive scalar particle. States are momentum-space wavefunctions on the mass shell $p^2 = -m^2$, $p^0 > 0$:

$$\mathcal{H} = \left\{ \psi(\mathbf{p}) \,\middle|\, \int \frac{d^3p}{2E_p} |\psi(\mathbf{p})|^2 < \infty \right\}, \qquad E_p = \sqrt{\mathbf{p}^2 + m^2},$$
$$\langle \psi | \phi \rangle = \int \frac{d^3p}{2E_p}\, \psi^*(\mathbf{p})\, \phi(\mathbf{p}).$$

The measure $d^3p/2E_p$ is Lorentz invariant.

Action of the Poincaré transformations.

- **Translations**: $U(a)\psi(p) = e^{-ip\cdot a}\psi(p)$, with $p \cdot a = -p^0 a^0 + \mathbf{p}\cdot\mathbf{a}$.
- **Lorentz transformations**: $U(\Lambda)\psi(p) = \psi(\Lambda^{-1}p)$.

Unitarity. For translations the phases cancel,

$$\langle U(a)\psi | U(a)\phi \rangle = \int \frac{d^3p}{2E_p}\, e^{+ip\cdot a}\psi^*(p)\, e^{-ip\cdot a}\phi(p) = \langle \psi | \phi \rangle,$$

and for Lorentz transformations the change of variables $p' = \Lambda^{-1}p$ leaves the invariant measure unchanged:

$$\langle U(\Lambda)\psi | U(\Lambda)\phi \rangle = \int \frac{d^3p'}{2E_{p'}}\, \psi^*(p')\, \phi(p') = \langle \psi | \phi \rangle.$$

The Klein–Gordon Inner Product

For solutions of the Klein–Gordon equation the inner product is defined on a spacelike hypersurface Σ, for example $t = 0$:

$$\langle \phi, \psi \rangle = i \int_\Sigma d^3x \, [\phi^* \partial_t \psi - (\partial_t \phi^*)\psi] \, .$$

It is the flux through Σ of the conserved current $j^\mu = -i(\phi^* \partial^\mu \psi - \partial^\mu \phi^* \, \psi)$ (the minus sign because $\partial^0 = -\partial_t$ in our metric, so that $\langle \phi, \psi \rangle = \int_\Sigma d^3x \, j^0$). Because $\partial_\mu j^\mu = 0$ for solutions, the integral is the same on every spacelike hypersurface, so it is **Lorentz invariant**: evaluating it on the boosted surface $t' = 0$ gives the same number. It is **positive** on positive-frequency solutions and **indefinite** on general ones; that indefiniteness, not a failure of invariance, is what stops the single-particle Klein–Gordon theory from having a probability interpretation, and it is resolved by restricting to positive frequencies, which is the Hilbert space above. Defining the inner product on spacelike hypersurfaces rather than over all of spacetime ensures

- **Causality**: data on Σ determine the evolution inside the light cones.
- **Unitarity**: probability is conserved, via the conserved current j^μ.
- **A well-posed initial-value problem**: hyperbolic equations such as the Klein–Gordon equation need data on Σ.